

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	01 October 2026
Team ID	SWTID-2026-4498
Project Name	Import data using Transform Maps
Maximum Marks	4 Marks

Technical Architecture:

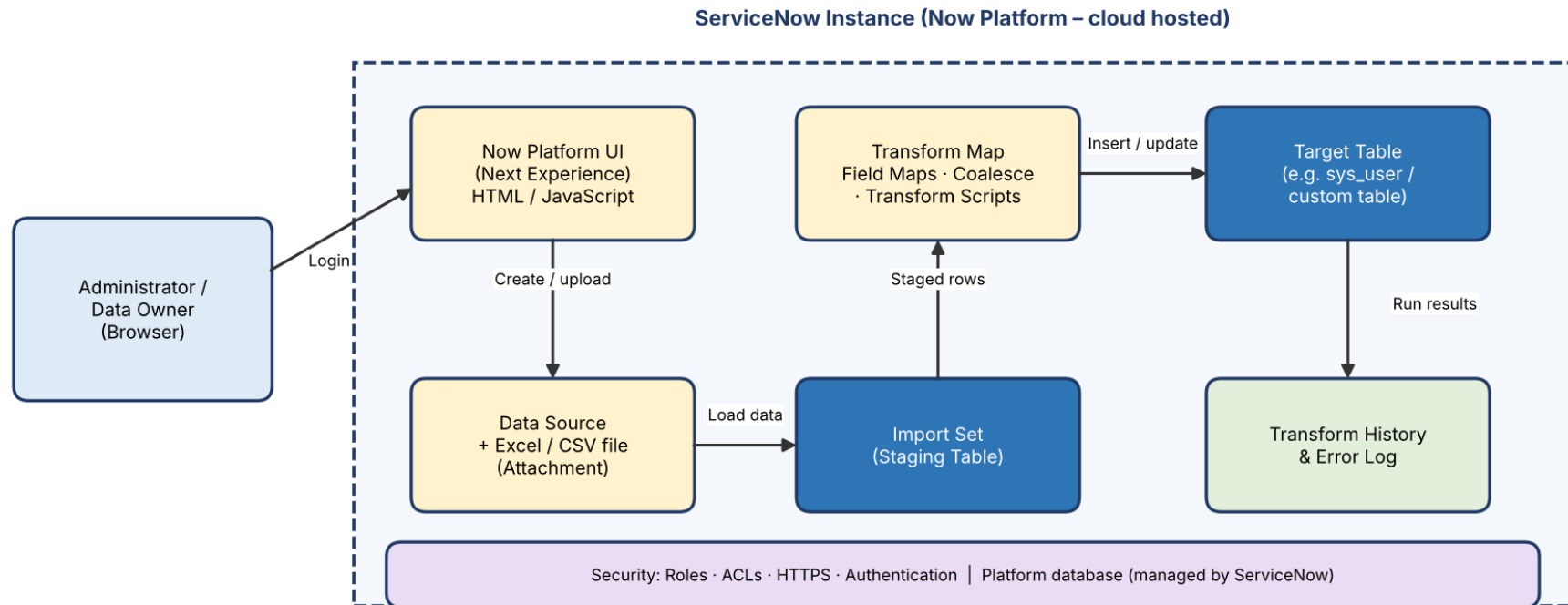


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Administrator and data owner use the browser-based ServiceNow interface to create the data source, import set and Transform Map.	ServiceNow Now Platform UI (Next Experience), HTML / JavaScript
2.	Application Logic-1	Data Source and Import Set: loads the Excel/CSV file rows into the staging (import set) table.	Data Source, Load Data, Import Set
3.	Application Logic-2	Transform Map: maps source fields to target table fields and uses a coalesce field to update existing records.	Transform Map, Field Maps, Auto Map Matching Fields
4.	Application Logic-3	Transform scripts and validation: clean or convert values and skip invalid rows during the transform.	JavaScript (Glide API) – onBefore / onAfter and field scripts
5.	Database	Import set staging table and target table (e.g., User / custom table) holding the imported records.	ServiceNow platform relational database (tables)
6.	Cloud Database	Instance and its database are hosted and managed by ServiceNow in the cloud.	ServiceNow Cloud (Now Platform)
7.	File Storage	Source Excel/CSV file is stored as an attachment on the Data Source record.	ServiceNow Attachments (sys_attachment)
8.	External API-1	Not required; the data comes from an uploaded file. REST / JDBC data sources can be used for other sources.	None (optional REST / JDBC data source)
9.	External API-2	Not used in this project.	None
10.	Machine Learning Model	Not used in this project.	None
11.	Infrastructure (Server / Cloud)	Local Server Configuration: not required. Cloud Server Configuration: ServiceNow instance (Personal Developer Instance) hosted on ServiceNow cloud.	ServiceNow Cloud, Personal Developer Instance (PDI)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	ServiceNow is a proprietary low-code platform; the project uses standard open file formats and scripting.	CSV / Excel file formats, JavaScript
2.	Security Implementations	Role-based access (admin / import_admin), ACLs on import tables, secure login and HTTPS access to the instance, import actions logged.	Roles, ACLs, HTTPS/TLS, authentication
3.	Scalable Architecture	Multi-tier cloud platform (UI, application and database tiers); staging table separates loading from transforming so larger files and scheduled imports can be handled.	ServiceNow Now Platform, Import Sets, Scheduled Imports
4.	Availability	Instance runs on ServiceNow's managed cloud with redundant infrastructure and backups; imports can be re-run safely using the coalesce field.	ServiceNow cloud hosting, coalesce
5.	Performance	Records are loaded in bulk to a staging table, then transformed in batches; coalesce on an indexed field and efficient transform scripts keep imports fast.	Import Set staging, Transform Map, indexed coalesce field